Final Seminar Script: Under 6 Minutes

Onedrive link to video: 20260510_223531000_iOS.MOV

Scene 1: Hook and clinical gap

Approx. 40 seconds

In *The Empire Strikes Back*, Luke Skywalker receives a prosthetic hand that can feel.

That scene is fiction, but the goal is real.

For millions of people who rely on prosthetic limbs, the challenge is not only restoring movement. It is restoring a sense of touch that feels useful, reliable, and natural.

Modern upper-limb prostheses can restore some movement, but they still struggle to restore sensory feedback.

A conventional myoelectric prosthesis is mostly one-way. The user contracts a residual muscle, surface electrodes record EMG from the skin, and that signal is decoded to control movement.

A closed-loop system adds the missing return pathway: stimulation can return sensory information back to the nervous system.

That is the promise: movement plus sensation.

But this promise creates a problem.

The same stimulation used to restore sensation can corrupt the EMG signal needed for control.

Scene 2: The core technical problem

Approx. 30 seconds

The technical problem is stimulation artefact.

When stimulation is delivered, it creates a large voltage transient in the recording channel. This artefact can be much larger than the underlying EMG signal.

So the recorded signal is no longer just muscle activity. It becomes EMG plus stimulation artefact, plus evoked muscle response, plus noise.

That matters because the controller does not know which part of the signal is useful control information and which part is contamination.

So the central question of this thesis is:

How can EMG remain usable when stimulation artefacts enter the same recording loop?

Scene 3: Current fixes and literature gap

Approx. 40 seconds

Several approaches already exist.

Blanking ignores the recording during and after each stimulation pulse. It is fast, but it throws data away.

Template subtraction estimates a repeated artefact shape and subtracts it. This can work when the artefact is stable, but real artefacts can change over time.

Adaptive and reference-based methods try to track or separate the artefact more intelligently, but their performance depends on reference quality, convergence, computational cost, and how predictable the artefact remains.

So the gap is not that no methods exist.

The gap is that these methods are often tested on different datasets, under different assumptions, and using different metrics.

That makes it hard to know which method is actually robust when conditions become harder.

Scene 4: Aims

Approx. 25 seconds

This thesis had three aims.

First, to build a fair benchmark where every algorithm is tested on the same data using the same metrics.

Second, to stress-test those algorithms under escalating artefact conditions.

Third, to identify the most robust real-time candidate, meaning the method most worth taking forward into hardware validation.

In simple terms:

Which artefact suppression method deserves to be tested next on real hardware?

Scene 5: Approach and pipeline

Approx. 50 seconds

To build the benchmark, I used clean EMG from NinaPro DB3.

This dataset includes 11 transradial amputees, 49 hand movements plus rest, 12 Delsys Trigno electrodes, and a native sampling rate of 2 kilohertz.

The baseline EMG was real. The contamination was synthetic.

That was important because it gave the benchmark a known ground truth. I could start with clean EMG, inject controlled artefacts, apply each suppression method, and compare the recovered output against the original clean signal.

The same pipeline was used for every method.

Clean EMG was combined with controlled stimulation artefacts and M-wave components. The corrupted signals were passed through nine suppression algorithms, then evaluated using four metrics:

RMSE, Pearson correlation, SNR, and latency.

Together, these metrics tested both signal quality and real-time feasibility.

Scene 6: Stress sweep

Approx. 40 seconds

The key methodological decision was the stress sweep.

Instead of testing each algorithm under one ideal condition, I tested every method under three escalating conditions: Static, Moderate, and Stress.

Static represents the ideal case, where the artefact and M-wave are perfectly repeatable.

Moderate adds smaller dynamic changes: timing jitter, fatigue variation, morphology variation, and decay variation.

Stress increases those same mechanisms.

These levels were not intended to copy one clinical protocol exactly. They were designed as a controlled hierarchy of difficulty.

The important point is that the data source, artefact framework, algorithms, and metrics stayed constant.

Only the difficulty changed.

That is what makes the comparison fair.

Scene 7: Results

Approx. 65 seconds

The results show a clear robustness gap.

Across the stress sweep, most methods degraded as the artefact became more dynamic. This means that methods which looked acceptable under simpler conditions did not necessarily remain reliable when timing, amplitude, shape, and decay began to vary.

GSO was the clearest exception.

Across all three conditions, GSO remained the most stable real-time candidate. Its SNR stayed around 5 decibels, its RMSE remained close to 122 to 125 microvolts, its Pearson correlation stayed around 0.82 to 0.84, and its latency was low, at about 1.4 milliseconds per cycle.

The reason GSO performed well is structural.

It does not rely on memorising one fixed artefact waveform. Instead, Gram-Schmidt orthogonalisation removes the artefact-related subspace from the recording.

So even when the artefact changes, GSO can still suppress the broader artefact structure.

In this benchmark, GSO was therefore the strongest candidate to take forward into hardware validation.

Scene 8: Significance and conclusion

Approx. 45 seconds

The contribution of this thesis is not a finished prosthetic system.

The contribution is a reproducible evaluation pipeline for stimulation artefact suppression in closed-loop EMG control.

I generated EMG signals with controlled artefact contamination, tested nine suppression methods under the same conditions, and evaluated the recovered signals using SNR, RMSE, Pearson correlation, and latency.

That means this benchmark can be used as a method-selection tool.

It shows which methods remain stable, which ones degrade, and which candidate is most worth taking into hardware testing.

Within this framework, GSO was the most robust real-time candidate.

So the next step is not to claim clinical readiness.

The next step is to implement and validate GSO on real recording hardware with real stimulation artefacts.

Luke’s hand was fiction.

But the engineering problem behind that scene is real.

And this thesis takes one step toward solving it.